

Project Title

Optional Subtitle



Notes / Thesis / Report

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Chapter 1

Introduction

“Everything should be made as simple as possible, but no simpler.”

This template is designed for longer mathematics or physics documents that mix structured exposition, reusable theorem environments, figures, exercises and bibliography support.

1.1 How to use it

Start by editing `metadata.tex`, then replace these starter chapters with your own material. The package setup is centralized in `style/project.sty`, while reusable macros live in `style/commands.tex`.

Definition 1.1.1: Template idea

A good project template keeps recurring setup in one place so that future documents can focus on content rather than package cleanup.

Every `\defn` title is added to the “Index of Definitions” at the end of the document automatically. Use `\defn[key]{Title}{...}` to index a different key, or `\defn[-]{Title}{...}` to skip the entry.

Remark.

If a future project needs fewer features, switch them off through the package options in `main.tex` rather than letting the preamble grow ad hoc.

(*Haïoun*) (`AmbrosinoMonodromydefectsChernSimonstheoryHolography2026`)

Chapter 2

Foundations

2.1 Basic math commands

Your standard shortcuts are already available:

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

2.2 Theorem environments

Each one comes in up to four flavours: plain, referenceable (`\r`), with an attached proof (`\p`), and both (`\pr`). In the `\paper` look they are plain `\amsthm` environments, in the `\colour` and `\mono` looks the same commands render as boxes.

Theorem 2.2.1: Sample theorem

Let V be a vector space over $\mathbb{R}$. Then $0 \cdot v = 0$ for all $v \in V$.

Proof:

This follows from distributivity:

$$0 \cdot v = (0 + 0) \cdot v = 0 \cdot v + 0 \cdot v.$$

Subtracting $0 \cdot v$ from both sides gives the claim. ■

All numbered environments share one counter, so refer to them with plain `\ref` and write the type word yourself: the statement above is Theorem 2.2.1.

Lemma 2.2.2

An empty title argument renders just “Lemma”.

Proof:

Nothing to show. ■

Proposition 2.2.3

Propositions and corollaries carry no title at all.

Proof:

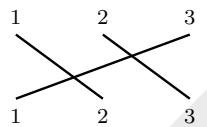
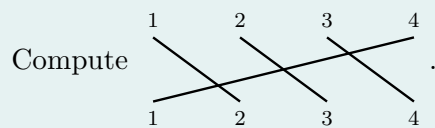
Their proof block is attached with `\propp` and `\corp`.

Fact 2.2.4

`\fact` is numbered, while `\factb` is an unnumbered aside like a remark.

Example.

The permutation helper from the notes project is also included:

**2.3 Exercises****Exercise 2.3.1: Permutations****Exercise 2.3.2: Exercise with an answer**

State the order of the cycle (123) .

Answer:

It is 3, since applying the cycle three times returns every element to its starting position.

Chapter 3

Results and references

3.1 Citations

By default the template uses `bibtex` with the `biber` backend and prints author-only citations: `\cite{key}` gives *Author*, `\citep{key}` gives (*Author*), and a postnote is added with `\cite[Chapter~5]{key}` *Author, Chapter 5*.

Entries live in `ref/references.bib`. Loading the template with the `[bibtex]` option switches back to `natbib` with the style named in `metadata.tex`.

3.2 Figures

Figures are looked up in `figures/` automatically.



Figure 3.1: Placeholder figure area for future projects.

3.3 Draft mode

Loading the template with `[draft]` puts a light watermark behind the text, adds a footer with the compilation date, and reveals notes to self: **[TODO: write the conclusion]** — the same note prints nothing in the final version. This sentence only exists in the draft.

Bibliography

Author, Alice (2024). *A Sample Reference*. Example Press.

Haïoun, Benjamin (2026). *Defining Extended TQFTs via Handle Attachments*. DOI: 10.48550/arXiv.2412.14649. arXiv: 2412.14649 [math.GT]. URL: <http://arxiv.org/abs/2412.14649> (visited on 07/22/2026).

Index of Definitions

Definitions are collected automatically from the definition boxes.

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